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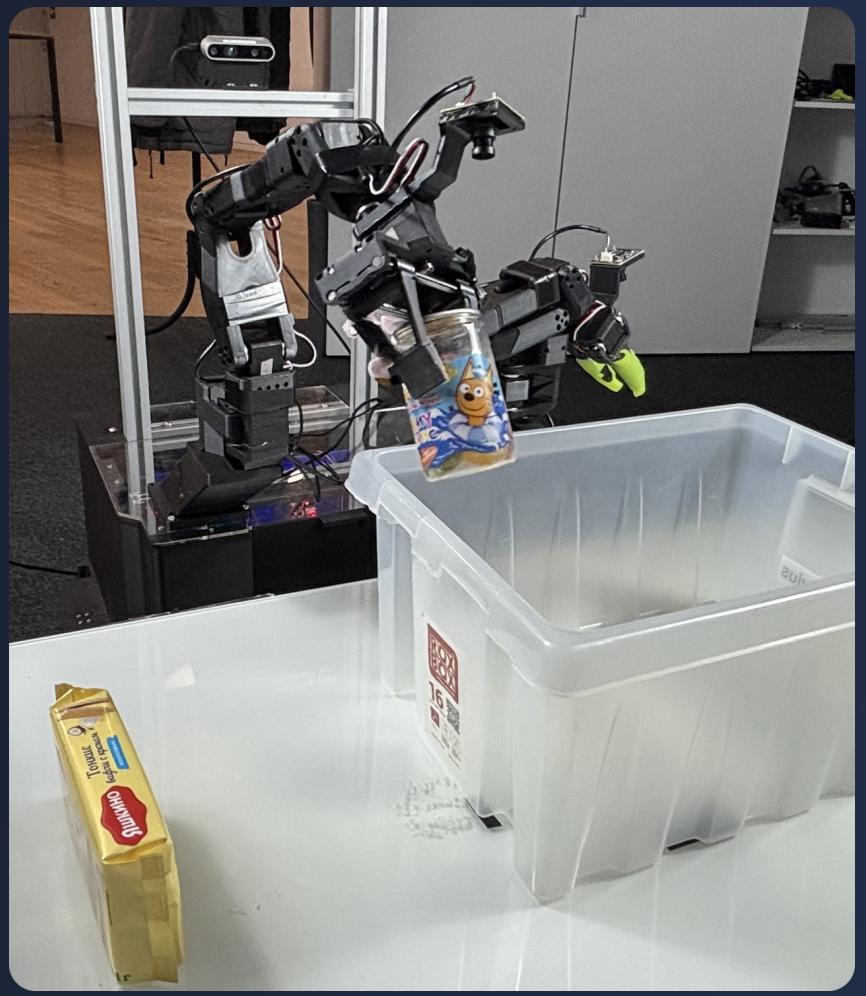
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HapticVLA

Contact-Rich Manipulation via Vision-Language-Action Model without Inference-Time Tactile Sensing

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Intelligent Space Robotics Laboratory, Skoltech · *equal contribution



arXiv:2603.15257

Robots see. They don't *feel.*

Camera-only VLAs

- ✓ See where the egg is
- ✗ Can't feel how hard they squeeze

Crushed eggs, crumbled waffles

Tactile sensors

- ✓ Feel every contact
- ✗ Costly, and tied to special hardware

Hard to put on every robot

What if a robot could learn the feel... *and leave the sensors behind?*

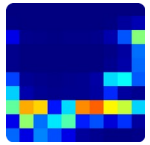
T O U C H

Touch in *training.* Not in deployment.

TRAINING · IN THE LAB

The robot can feel

Vision | Language | **Touch**



200 taxels on two fingertips score every grasp

DEPLOYMENT · ANYWHERE

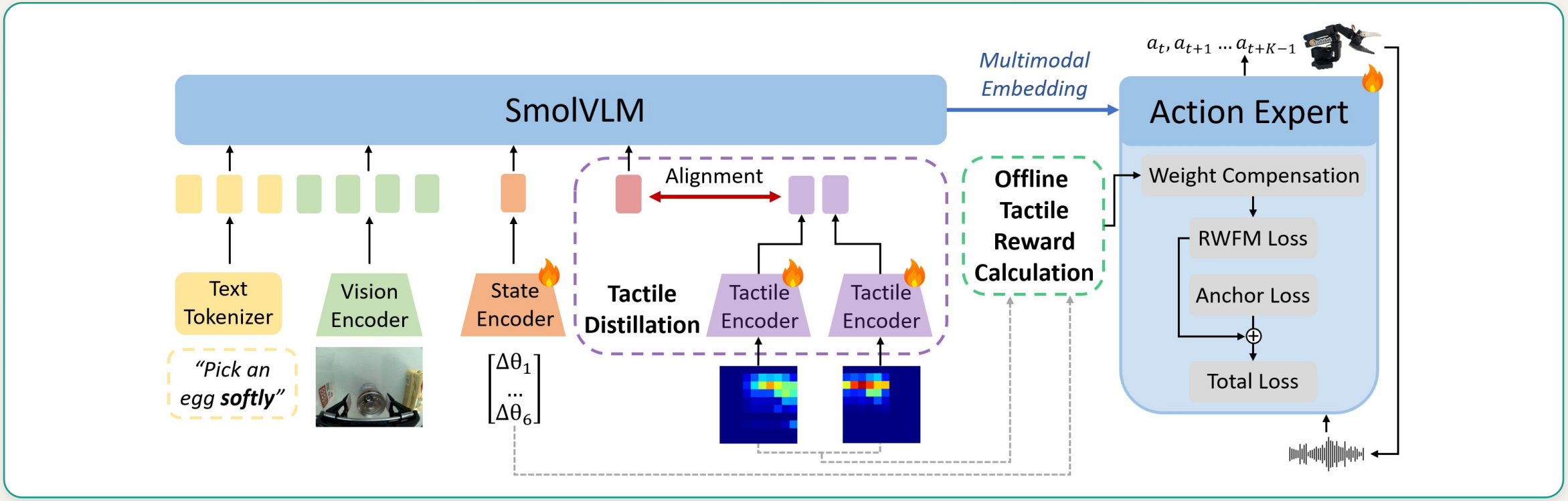
Cameras are enough

Vision | Language | No touch

Any camera-only robot runs the same small model

Touch becomes a teacher, not a part you have to install.

Use touch as a *reward.*



1 **Score.** Fingertips rate every moment of every demo.

2 **Penalize.** Over-force, slip, unsafe pressure lose weight.

3 **Reward.** Gentle, successful moves count more.

Cut the sensors. It gets *better.*

TEACHER

SmolVLA + touch

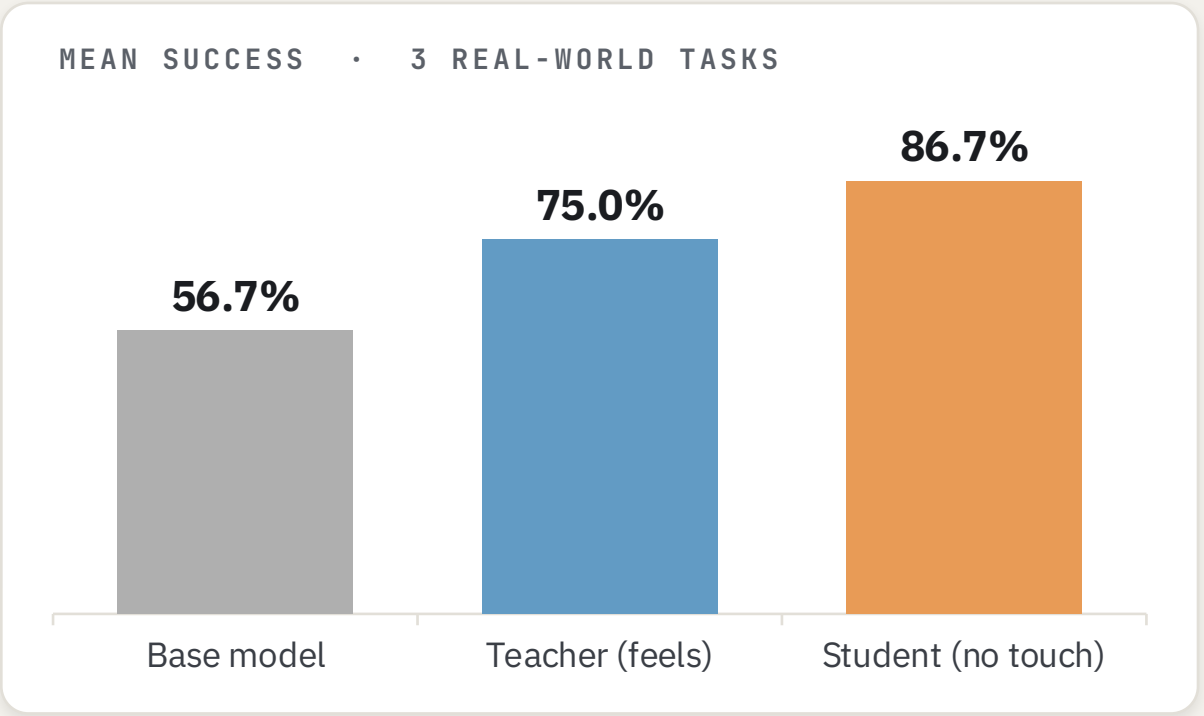
Feels. Labels all the data once, offline.

learns from 50 / 50: human demos + teacher actions

STUDENT = HapticVLA

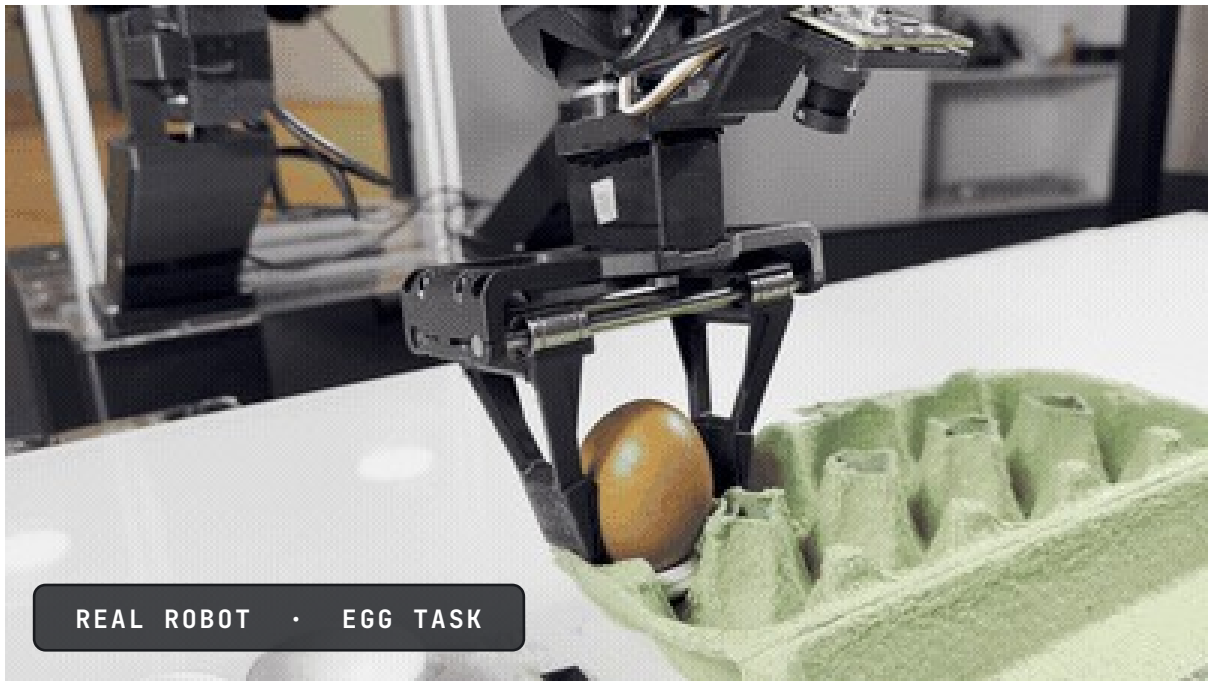
Same SmolVLA, no touch

Cameras + joint angles only. Still 0.45B.



The student that can't feel *beats* the teacher that can. 86.7% vs 75.0%

Hold the *egg.*





Jar

Deformable, keep its shape

50% → 75%



Waffles

Fragile, no cracks or crumbs

70% → 90%



Egg

Out of the carton, onto the tray

50% → 95%

86.7% HapticVLA, mean success

+30 pts vs. the base model

0 tactile sensors at deployment

20 real-robot trials per task · success = object intact and on target

THE TAKEAWAY

Train with touch. Deploy without it.

Touch can be a training-time modality. The distilled policy runs on any camera-only robot, at no extra inference cost.

Come see the robot at our poster

Project page

Video

Code + data

PAPER
arXiv:2603.15257

CODE + DATASET + DIGITAL TWIN
github.com/Advanced-Robotic-Manipulation/crab

MODELS
huggingface.co/armteam

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